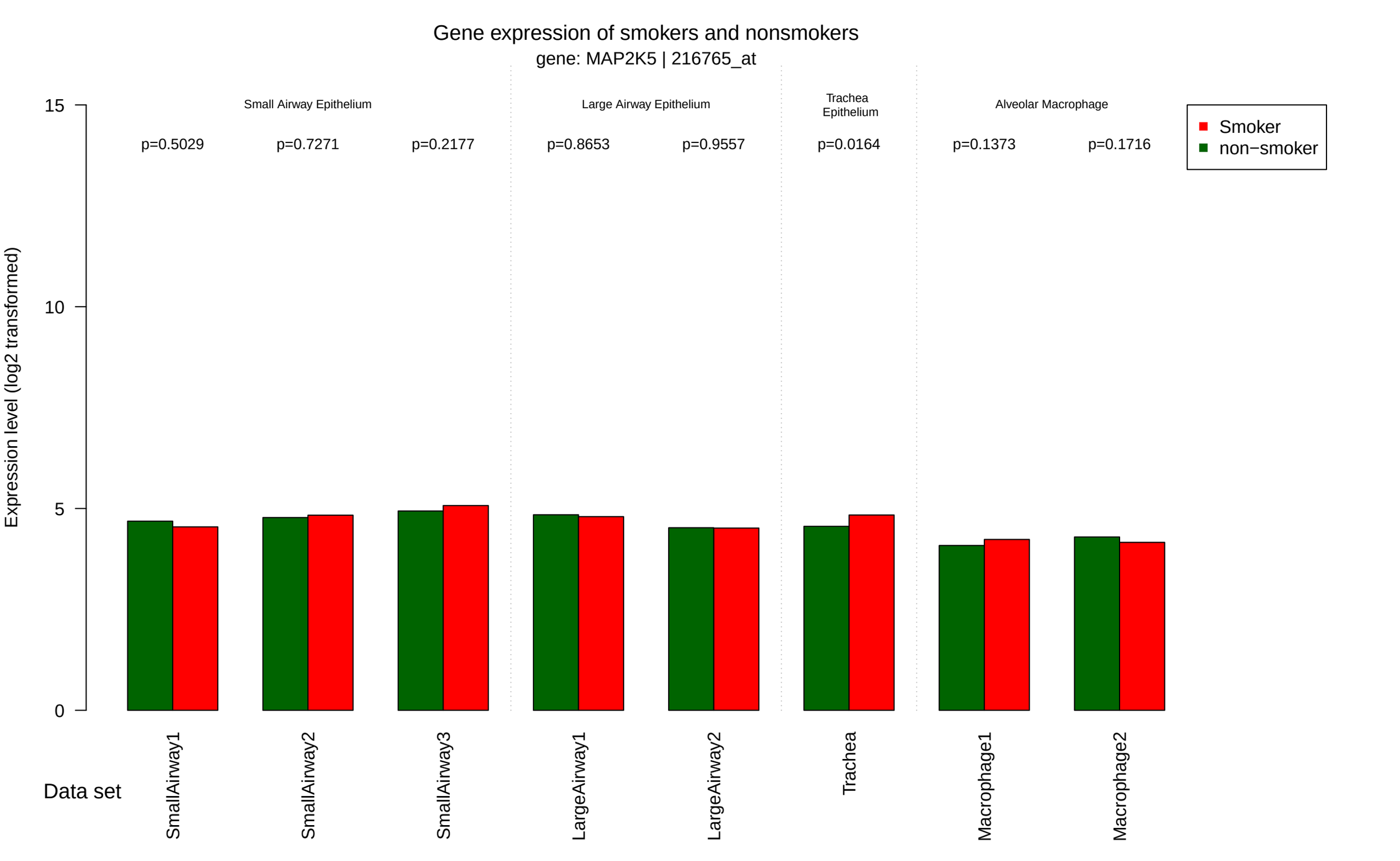
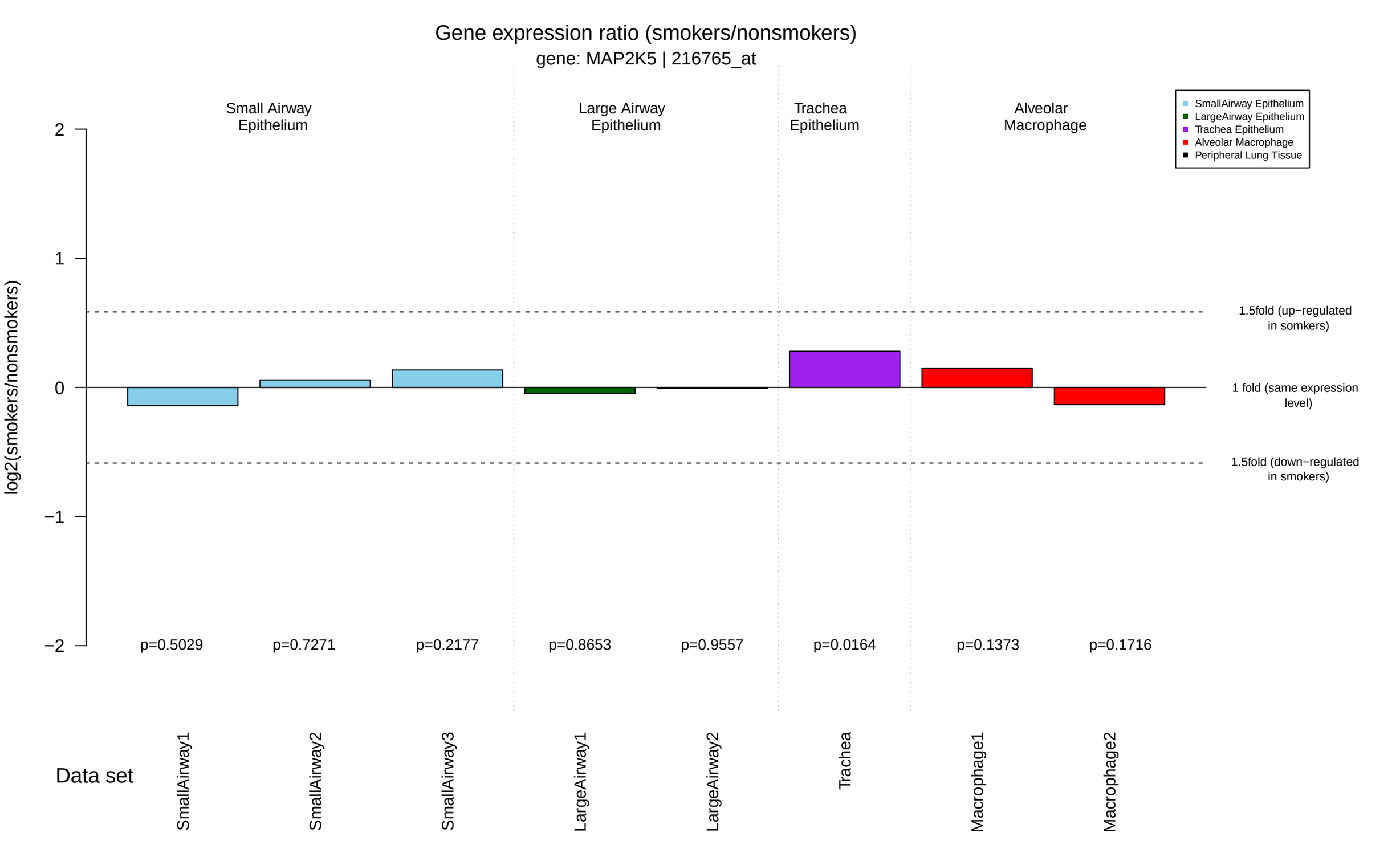
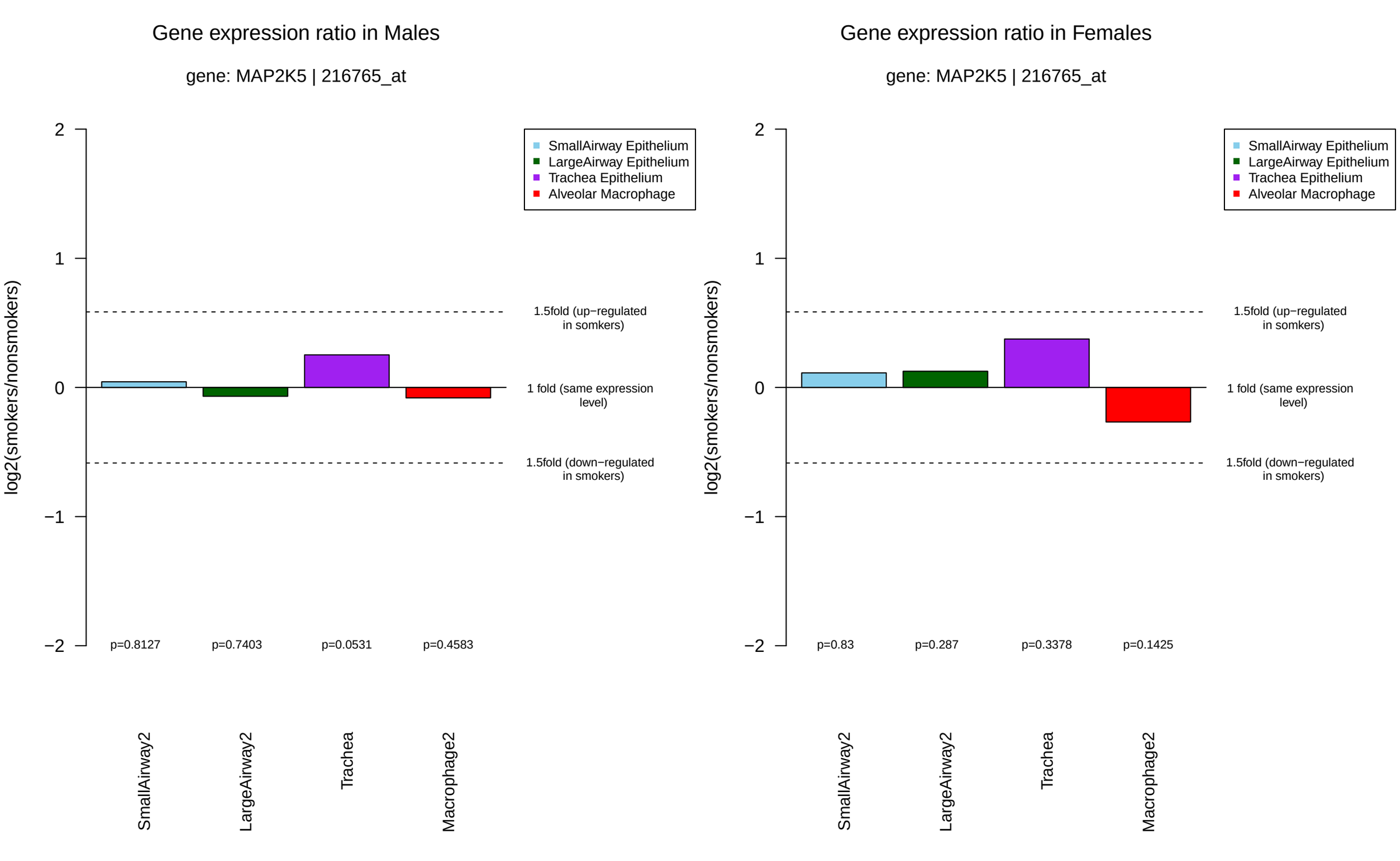
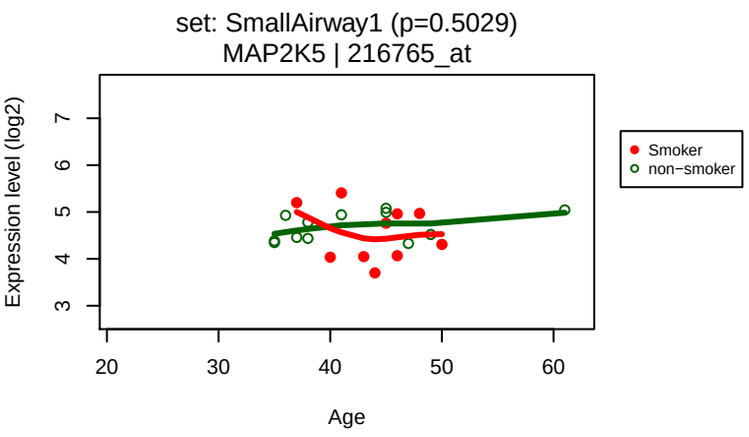
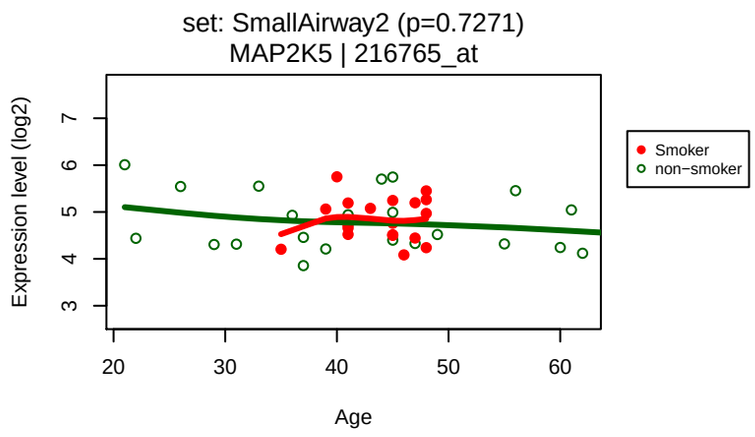
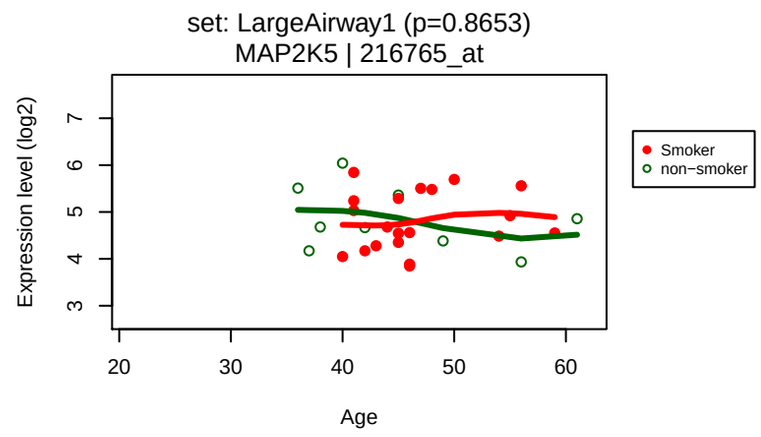

Gene expression of smokers and nonsmokers
gene: MAP2K5 | 216765_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5029
p=0.7271
p=0.2177
p=0.8653
p=0.9557
p=0.0164
p=0.1373
p=0.1716
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

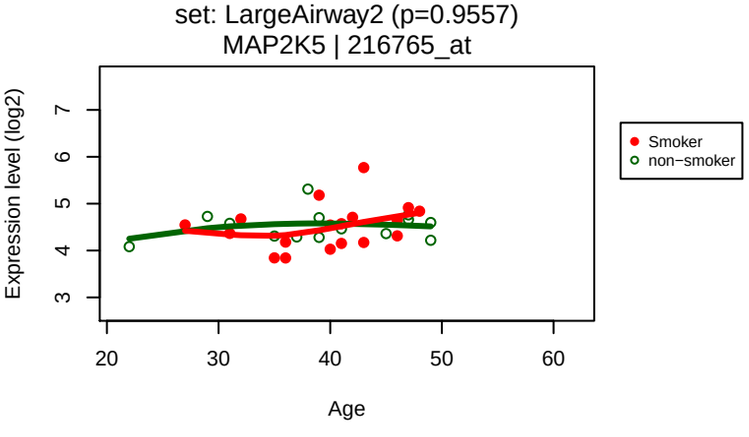

Gene expression ratio (smokers/nonsmokers)
gene: MAP2K5 | 216765_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5029
p=0.7271
p=0.2177
p=0.8653
p=0.9557
p=0.0164
p=0.1373
p=0.1716
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

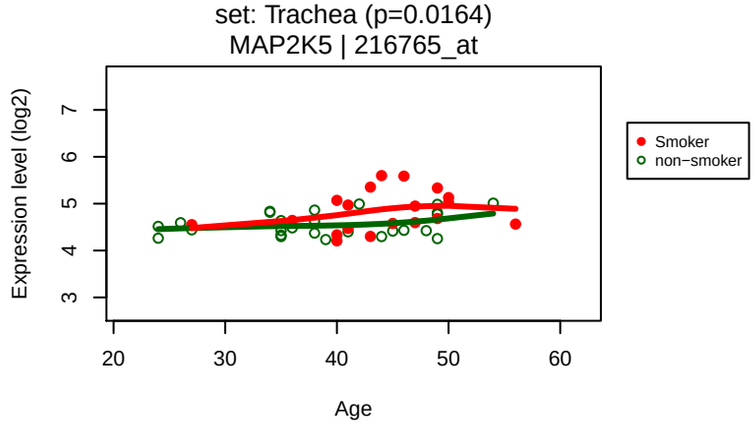

Gene expression ratio in Males
gene: MAP2K5 | 216765_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8127
p=0.7403
p=0.0531
p=0.4583
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MAP2K5 | 216765_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.83
p=0.287
p=0.3378
p=0.1425
SmallAirway2
LargeAirway2
Trachea
Macrophage2

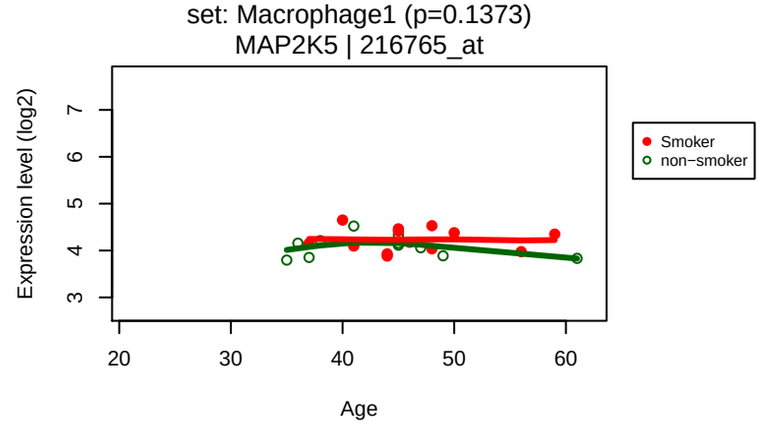

set: SmallAirway1 (p=0.5029)
MAP2K5 | 216765_at
Expression level (log2)
Age
Smoker
non−smoker

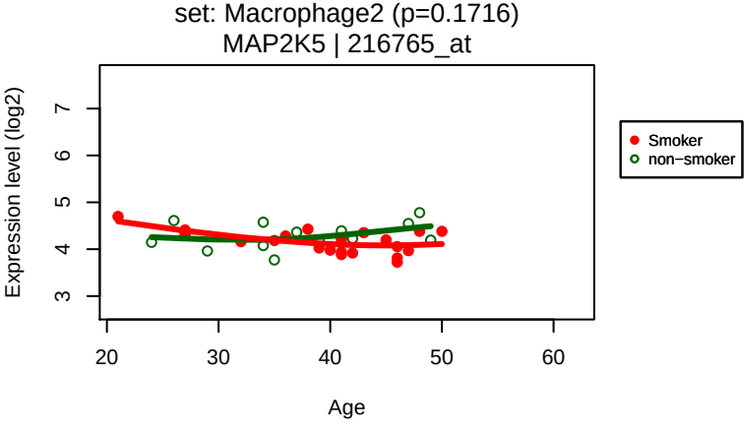

set: SmallAirway2 (p=0.7271)
MAP2K5 | 216765_at
Expression level (log2)
Age
Smoker
non−smoker

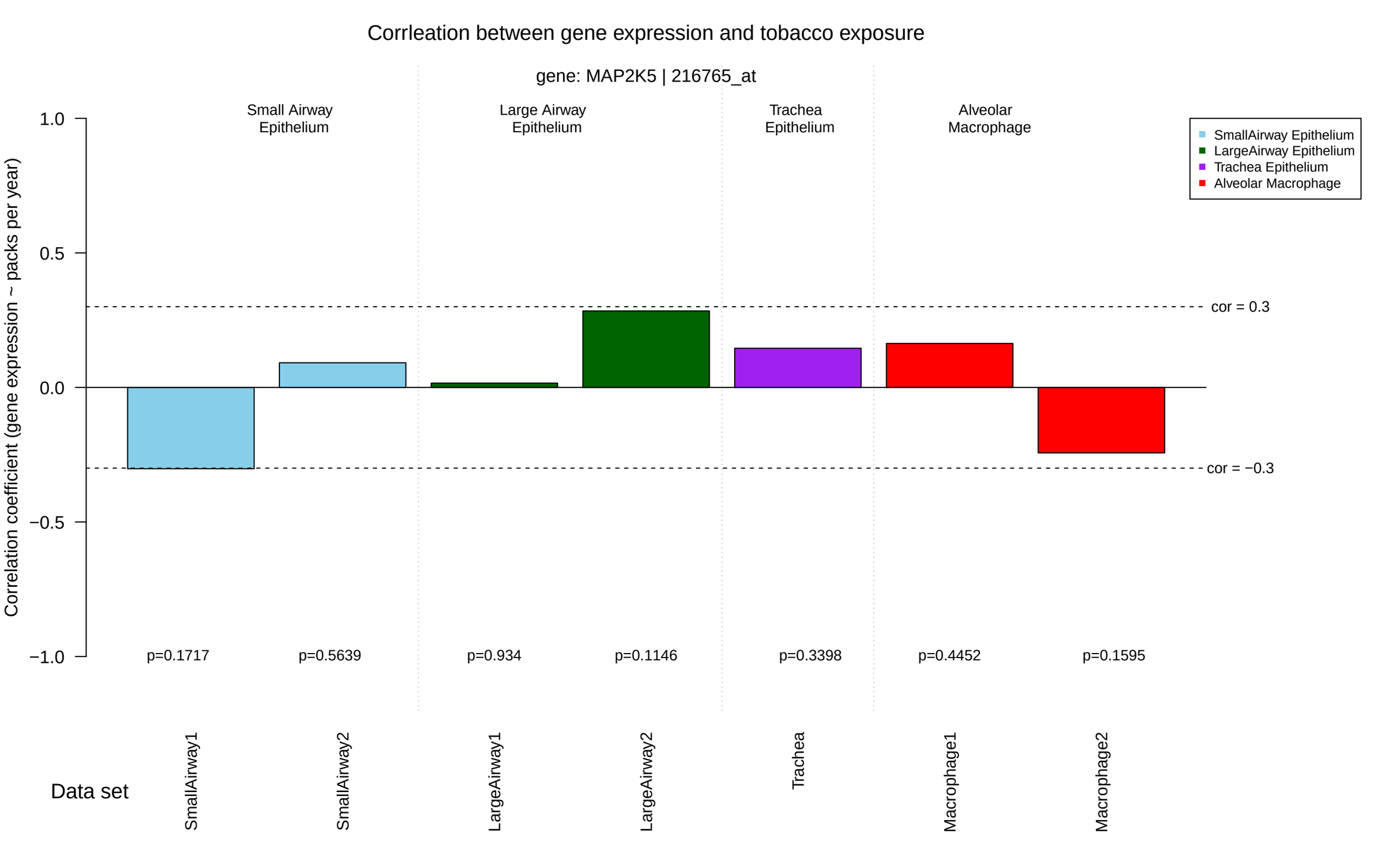

set: LargeAirway1 (p=0.8653)
MAP2K5 | 216765_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9557)
MAP2K5 | 216765_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0164)
MAP2K5 | 216765_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1373)
MAP2K5 | 216765_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1716)
MAP2K5 | 216765_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MAP2K5 | 216765_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1717
p=0.5639
p=0.934
p=0.1146
p=0.3398
p=0.4452
p=0.1595
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2